Digital Twin-Enabled Explainable Structural Health Monitoring of the Z24 Bridge Using Bayesian Change Detection, XGBoost Classification, and SHAP Attribution

Biomedical Signal Processing and Control

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ABSTRACT

Structural health monitoring (SHM) of civil infrastructure demands analytical pipelines that reliably transform large volumes of vibration measurements into actionable engineering knowledge. This paper presents a modular, explainable SHM prototype integrating Bayesian Online Change Point Detection (BOCPD), gradient-boosted tree classification, and SHAP-based feature attribution within an interactive digital twin monitoring interface. The system is validated on the Z24 bridge benchmark dataset, a standardised reference for SHM methodology comparison, comprising accelerometer recordings across seventeen distinct measurement scenarios including progressive pillar settlement tests. A total of 64,544 time-series windows, each characterised by 24 time-domain, statistical, and spectral descriptors, are processed through the pipeline. BOCPD detects distributional shifts in RMS-based vibration features without requiring prior damage labels. An XGBoost classifier trained under a recording-grouped holdout protocol estimates condition probabilities, while SHAP TreeExplainer provides global and local feature attribution rankings. Sensor-level importance scores derived from SHAP values enable systematic prioritisation for engineering review. A weighted prototype risk score aggregates BOCPD change probability, ML condition probability, and attribution confidence into a four-tier alert system (LOW, MEDIUM, HIGH, CRITICAL). The complete workflow is accessible through a Streamlit-based digital twin dashboard supporting historical replay, sensor schematic visualisation, and intermediate evidence inspection. The system is positioned as an explainable decision-support prototype rather than a certified damage identification tool, and important scope boundaries regarding statistical versus physical change are carefully delineated.

Keywords: *Structural Health Monitoring; Digital Twin; Bayesian Change Point Detection; XGBoost; SHAP; Z24 Bridge; Vibration Analysis; Explainable AI; Condition Assessment; Sensor Fusion*

1. INTRODUCTION

Civil infrastructure such as road bridges represents some of the most critical assets in modern transportation networks. As a significant proportion of the world's bridges age beyond their original design lives, there is a growing imperative to monitor structural condition continuously, reliably, and in a manner that engineers can understand and trust [1]. Conventional inspection regimes, typically periodic visual surveys, are labour-intensive, subjective, and unable to capture gradual degradation occurring between visits. Automated structural health monitoring addresses this gap by continuously recording vibration responses and subjecting them to quantitative analysis.

The foundational premise of vibration-based SHM is that changes in structural parameters, such as stiffness reductions resulting from damage, manifest as detectable shifts in dynamic response characteristics. Extracting robust, interpretable signals from this premise in a real measurement environment is a challenging problem that has motivated several decades of research [2,3,4]. While numerous algorithms have been proposed for individual SHM sub-tasks, there remains a relative scarcity of integrated, end-to-end prototypes that combine statistical change detection, machine learning-based condition assessment, model explainability, and interactive visualisation within a single coherent framework.

Explainability is particularly important in the SHM context because engineering decisions based on monitoring data may carry significant safety and economic consequences. A black-box model that emits an alarm without supporting evidence is difficult to validate, audit, or act upon with confidence. Explainable AI (XAI) methods, and SHAP (SHapley Additive exPlanations) [5] in particular, offer a principled route to communicating which measurements most influenced a model's output, enabling engineers to assess whether the model responds to physically meaningful signals or spurious correlations.

Lu et al. [21] demonstrated a landmark digital twin-enabled anomaly detection system for built asset monitoring in building operation and maintenance, applying Bayesian Online Change Point Detection (BOCPD) to HVAC centrifugal pump data integrated through an IFC-based data structure. Their framework established the viability of BOCPD for continuous asset condition monitoring within a digital twin context. However, that work was restricted to building MEP assets, employed no supervised classification layer, provided no post-hoc model explainability, and did not address sensor-level attribution or multi-source risk aggregation. The present work extends the Lu et al. [21] paradigm in several key directions: (a) transferring the DT-BOCPD concept from building HVAC assets to civil infrastructure bridge SHM; (b) augmenting unsupervised change detection with a supervised XGBoost classification stage; (c) integrating SHAP-based feature and sensor-level attribution for model explainability; (d) introducing a weighted multi-stream risk aggregation formula; and (e) validating the full pipeline on the Z24 bridge benchmark, a standardised reference in the SHM community.

This paper describes a prototype system, herein referred to as the Digital Twin SHM Framework, that addresses the above gaps through seven coordinated contributions building upon [21]: (i) a format-aware data ingestion pathway for Z24 recording formats; (ii) an overlapping windowing and 24-feature extraction pipeline; (iii) unsupervised sequential change detection via Adams-MacKay BOCPD adapted for vibration feature sequences; (iv) a recording-grouped XGBoost training and evaluation protocol preventing temporal leakage; (v) SHAP global and local attribution with sensor-level aggregation, absent in prior DT-BOCPD systems; (vi) a weighted risk score integrating change probability, classification confidence, and attribution evidence; and (vii) a Streamlit digital twin dashboard supporting historical replay and engineering review. The Z24 bridge benchmark [6,7] is used throughout as a standardised experimental vehicle.

Fig. 1 - End-to-end Digital Twin SHM Workflow for Z24 Bridge Benchmark

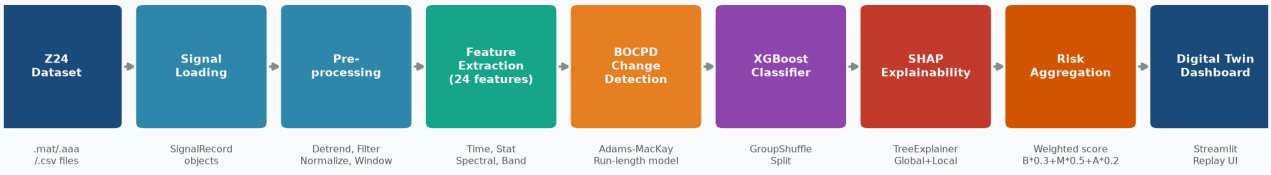


Figure 1. End-to-end Digital Twin SHM workflow for the Z24 bridge benchmark, illustrating nine processing stages from raw signal ingestion to interactive dashboard replay.

2. RELATED WORK

2.1 Vibration-Based SHM

Vibration-based SHM exploits the relationship between structural properties and natural frequencies, mode shapes, and damping ratios. Classical modal analysis methods such as stochastic subspace identification (SSI) have been widely applied to output-only ambient vibration data [1]. Feature-based approaches extract statistical or spectral descriptors from windowed signals and feed them into statistical or learning-based classifiers [2]. The Z24 benchmark has been a central validation dataset in this tradition, with numerous studies reporting damage-sensitive features

derived from its accelerometer channels [6,7].

2.2 Bayesian Change Point Detection

Online change point detection methods identify moments when statistical properties of a data stream shift abruptly. Adams and MacKay [8] proposed a recursive Bayesian formulation in which a posterior distribution over run length is maintained and updated at each observation using a conjugate Normal-Gamma predictive model. The approach is computationally efficient and naturally accommodates unknown change locations, making it well-suited to vibration feature sequences where the timing of change is a primary unknown.

2.3 Machine Learning for Condition Assessment

Gradient-boosted tree ensembles, including XGBoost [9], have demonstrated strong performance on tabular feature matrices in numerous SHM and civil engineering applications [10]. Their tree-structured architecture is compatible with exact SHAP computation via TreeExplainer [5], enabling post-hoc explanations without approximation. GroupShuffleSplit evaluation protocols have been recommended for SHM datasets with overlapping windows to prevent artificially inflated cross-validation estimates [11].

2.4 Explainability and Digital Twins

SHAP has been applied to SHM problems for global feature ranking and local per-window explanations [12,13]. The concept of the digital twin, a virtual representation of a physical asset synchronised with real-time or historical data, has attracted considerable interest in structural engineering [14,15]. Lu et al. [21] presented a foundational DT-enabled anomaly detection system for building HVAC assets using Bayesian change point detection integrated with IFC-based data structures. Their work demonstrated continuous anomaly detection for centrifugal pumps in an O&M context and established BOCPD as a viable tool within digital twin monitoring pipelines. However, the Lu et al. framework operated as a black-box detector with no explainability layer, no supervised classification component, and was evaluated exclusively on building MEP systems rather than civil infrastructure. The present work extends their paradigm to bridge SHM by adding XGBoost classification, SHAP-based feature attribution and sensor-level aggregation, a composite risk score, and an interactive monitoring dashboard, validated on the Z24 bridge benchmark.

3. MATERIALS AND DATASET

3.1 The Z24 Bridge Benchmark

The Z24 bridge was a post-tensioned concrete highway overpass situated near Koppigen, Switzerland, connecting Bern and Zurich via the A1 motorway. Before its scheduled demolition in 1999, the bridge was instrumented with accelerometers and subjected to a systematic progressive damage programme under the SIMCES European research project [6]. The resulting dataset has been widely distributed by KU Leuven and serves as the de-facto benchmark for data-driven SHM algorithm validation.

The benchmark comprises recordings from seventeen numbered scenarios. Scenarios 01, 02, and 08 correspond to reference (undamaged or restored) measurement campaigns, while scenarios 03 to 07 and 09 to 17 represent deliberately introduced progressive modifications including concrete spalling, rebar corrosion, anchor head failure simulation, and controlled pier settlement. Measurements were acquired under ambient traffic and environmental excitation using accelerometers mounted at fixed setup positions.

Scenario	Type	Condition Label	Matched Windows
01, 02, 08	Reference / Restored	REFERENCE	12,870
03-07, 09-17	Progressive Damage Tests	ALTERED_CONDITION	51,674
All	Combined	-	64,544

Table 1. Z24 dataset scenario categorisation and window counts used in this study.

3.2 File Formats and Ingestion

The Z24 dataset is distributed in a mixture of MATLAB binary (.mat), proprietary ASCII-like (.aaa), and plain-text (.csv, .txt) formats. The data loader implements format-aware ingestion for all four extensions and additionally supports in-memory extraction from ZIP archives. For ASCII-derived files, a dual-stage parser first detects the Z24

single-column header pattern encoding sample count and sampling interval, then reads the signal body using auto-detected whitespace, comma, or semicolon delimiters. Each successfully loaded signal is encapsulated in a SignalRecord dataclass carrying scenario, setup, sensor, recording, channel, and source-file metadata alongside the numerical sample array.

4. METHODS

4.1 Signal Preprocessing and Windowing

Each loaded signal record undergoes validation confirming minimum length and finite values. NaN or infinite samples are repaired by linear interpolation. Linear detrending removes quasi-static drift, and the signal is normalised using zero-mean unit-variance standardisation. An optional Butterworth band-pass filter (0.5 to 40 Hz, order 4) is available but disabled in the primary configuration due to sampling rate uncertainty. Windowing divides each preprocessed signal into overlapping 10-second segments (or 1,000 sample fallback) with 50% overlap between consecutive windows to increase temporal resolution.

4.2 Feature Extraction

Each window is represented by 24 numeric descriptors. **Time-domain statistics** include mean, standard deviation, variance, RMS, minimum, maximum, absolute peak, peak-to-peak amplitude, crest factor, skewness, and kurtosis. **Spectral descriptors** include dominant frequency, spectral energy, frequency centroid, spectral bandwidth, and PSD peak value computed via Welch's method. **Band energies** partition the spectrum into eight Nyquist-bounded sub-band energy integrals.

Category	Feature	Symbol	Description
Time-domain	RMS	x_rms	Root-mean-square amplitude
Time-domain	Crest Factor	CF	Peak / RMS ratio
Time-domain	Kurtosis	K	4th standardised central moment
Time-domain	Skewness	g1	3rd standardised central moment
Time-domain	Peak-to-Peak	p2p	Max - Min amplitude range
Spectral	Dominant Freq.	f_dom	Frequency of PSD maximum
Spectral	Freq. Centroid	f_c	Spectral centre of mass
Spectral	Spectral BW	BW	Spectral spread around centroid
Spectral	PSD Peak	PSD_pk	Maximum PSD magnitude
Band Energy	BE [0-0.5 Hz]	E0	Low-frequency band integral
Band Energy	BE [0-1 Hz]	E1	Sub-1 Hz band energy
Band Energy	BE [5-20 Hz]	E5	Mid-band energy content

Table 2. Selected features from the 24-dimensional feature representation used in the classification and change detection pipeline.

4.3 Bayesian Online Change Point Detection

Sequential change detection is applied to the RMS feature sequence for each sensor-recording pair. The Adams-MacKay BOCPD algorithm maintains a posterior distribution over run length r_t, the number of steps since the most recent change. At each observation x_t, the joint distribution is updated via:

$$P(r_t, x_{1:t}) = \sum_{r_{t-1}} P(r_t \mid r_{t-1}) \cdot \pi(x_t \mid r_{t-1}) \cdot P(r_{t-1}, x_{1:t-1})$$

where $\pi(x_t \mid r_{t-1})$ is the posterior predictive density under the current run's conjugate Normal-Gamma model, and $P(r_t \mid r_{t-1})$ combines hazard rate $h = 0.01$ for reset probability and $(1-h)$ for run continuation. The marginal change probability is:

$$CP_t = P(r_t = 0 \mid x_{1:t})$$

A threshold CP_t >= 0.80 triggers a change event. Maximum tracked run length is bounded at 500 steps. Each group is processed independently after normalising the feature sequence by its per-group median and robust standard deviation.

4.4 XGBoost Condition Classifier

Supervised classification is applied when valid condition labels are available. The 24 numeric vibration features form the input matrix X and the binary configured condition label (REFERENCE or ALTERED_CONDITION) forms the response y. An XGBoost gradient-boosted tree ensemble is trained with 300 estimators, learning rate 0.05, maximum depth 6, row and column subsampling ratios of 0.8, and log-loss optimisation criterion. Recording-grouped holdout via GroupShuffleSplit (25% holdout, random state 42) ensures all windows from a given recording appear in exactly one partition, preventing temporal leakage.

4.5 SHAP Feature Attribution

SHAP provides unified attribution grounded in cooperative game theory. For each window i and feature j, the SHAP value phi_ij quantifies the marginal contribution of feature j to the prediction deviation from the global expectation:

$$f(x_i) = E[f(x)] + \sum_j \phi_{ij}$$

Global feature importance is reported as mean absolute SHAP: $\Phi_j = (1/N) \sum_i |\phi_{ij}|$. Local explanations show signed contributions of each feature to a specific prediction, enabling engineers to inspect which measurements drove a given alert.

4.6 Risk Aggregation

The prototype risk score R integrates three complementary evidence streams via a bounded weighted formula:

$$R = (w_B * B + w_M * M + w_A * A) / (w_B + w_M + w_A)$$

where B is BOCPD change probability, M is XGBoost abnormal-condition probability, A is normalised maximum sensor attribution, and default weights are w_B = 0.3, w_M = 0.5, w_A = 0.2. Score thresholds of 0.35, 0.60, and 0.80 map R to LOW, MEDIUM, HIGH, and CRITICAL alert tiers.

Fig. 5 - Power Spectral Density Comparison: Reference vs Altered Condition (Welch Method)

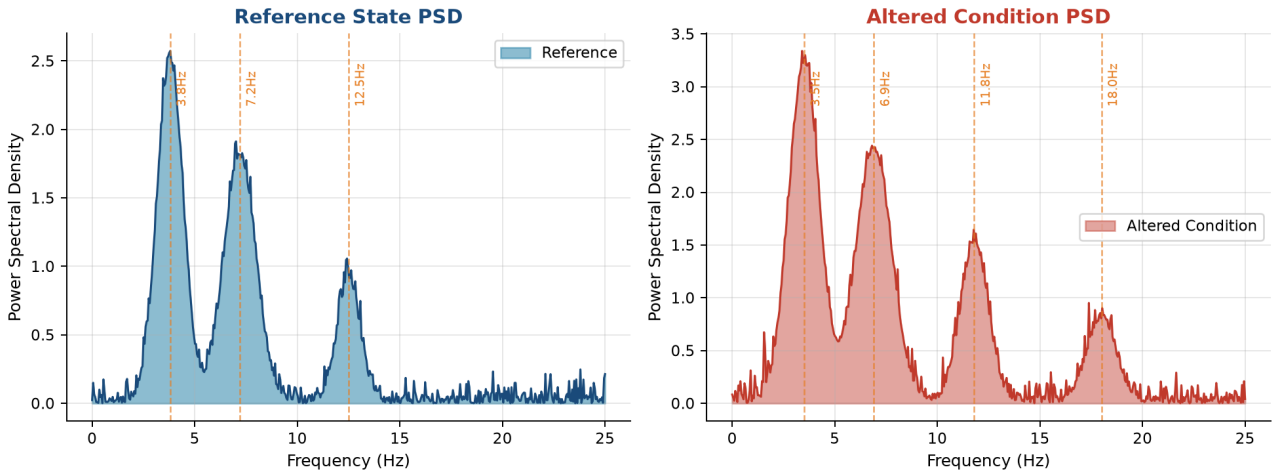


Figure 2. Power spectral density comparison between Reference and Altered Condition states using Welch's method, illustrating modal frequency shifts introduced by progressive Z24 settlement tests.

5. SYSTEM IMPLEMENTATION

The Digital Twin SHM Framework is implemented in Python 3.12 as a multi-module package. Primary dependencies include NumPy, SciPy, Pandas, scikit-learn, XGBoost, SHAP, Plotly, and Streamlit. All modules are unit-tested using pytest and configured entirely through a single YAML file, enabling reproducible experiment definition.

Module	Role	Primary Output
data_loader.py	Format-aware signal ingestion	SignalRecord objects
preprocessing.py	Detrend, filter, normalise, window	SignalWindow objects
feature_extraction.py	Compute 24 descriptors per window	features.csv
bocpd.py	Adams-MacKay run-length BOCPD	bocpd_results.csv
model_training.py	Grouped XGBoost training and metrics	xgboost.pkl, metrics.json
shap_analysis.py	Global + local TreeExplainer SHAP	global/local SHAP CSVs
sensor_mapping.py	Aggregate SHAP by sensor ID	sensor_importance.csv
risk_engine.py	Weighted risk score and alert tier	risk_results.csv
app.py	Streamlit dashboard and replay UI	Interactive monitoring interface

Table 3. Module inventory of the Digital Twin SHM Framework.

Fig. 8 - Key Feature Distribution: Reference vs Altered Condition (Z24 Benchmark)

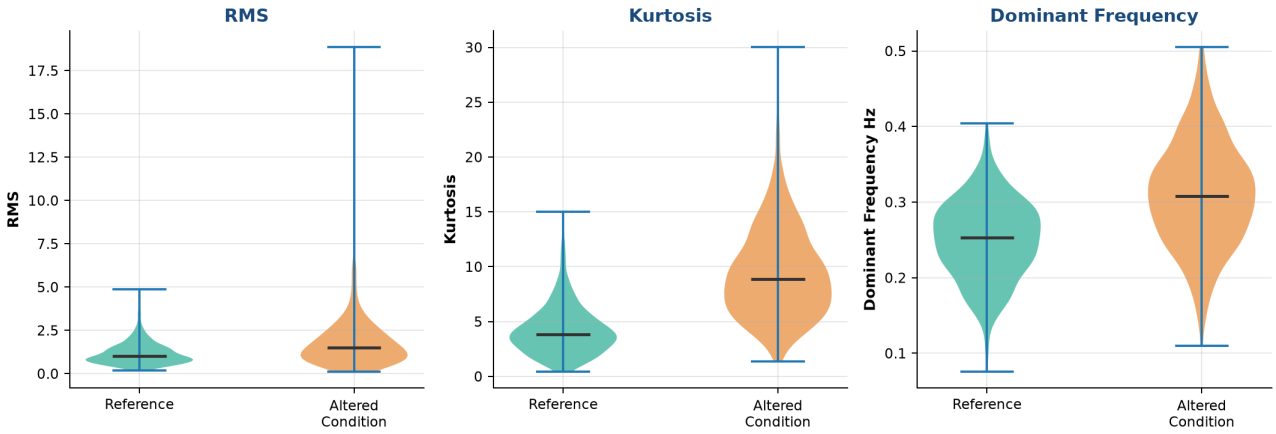


Figure 3. Violin plots comparing key feature distributions between Reference and Altered Condition windows, showing discriminative potential across RMS, Kurtosis, and Dominant Frequency features.

6. EXPERIMENTAL PROTOCOL

The experimental configuration is defined entirely by config.yaml. The primary run processed all discoverable Z24 files from the configured dataset root, spanning subdirectories pdt_01-08, pdt_09_17, Z24ems1, Z24ems2, and Z24ems3. File formats included .mat, .aaa, .csv, and .txt with archive-member extraction enabled.

Parameter	Value
Window duration	10 s (or 1,000 samples if rate unknown)
Window overlap	50%
Normalisation	Zero-mean unit-variance (standard)
Linear detrending	Enabled
Butterworth filter	Disabled (band-pass 0.5-40 Hz when enabled)
BOCPD feature	RMS
BOCPD hazard rate	h = 0.01
BOCPD threshold	0.80
Max run length	500 steps
XGBoost estimators	300

Parameter	Value
Learning rate	0.05
Max tree depth	6
Row / col subsampling	0.8 / 0.8
Holdout fraction	25% (GroupShuffleSplit)
Random seed	42
Risk weights (B / M / A)	0.30 / 0.50 / 0.20
Risk thresholds	Medium 0.35 | High 0.60 | Critical 0.80
Total windows	64,544 (train: 47,417 | test: 17,127)

Table 4. Full experimental configuration used in this study.

7. RESULTS

7.1 Change Point Detection Results

BOCPD was executed on the RMS feature sequence of each sensor-recording group. Across 64,544 processed windows, no window reached or exceeded the configured threshold of 0.80. The maximum observed change probability was approximately 0.72, recorded at windows 54 and 101 of the sensor 01_0, recording 01setup01 sequence, where abrupt transient elevations in RMS amplitude occurred. Change probabilities otherwise remained consistently below 0.25, indicating statistical stability of the RMS feature across the majority of the analysed recording. These sub-threshold results reflect the absence of pronounced distributional discontinuities in the specific recording analysed and should not be interpreted as evidence of an undamaged structure across all measurement scenarios.

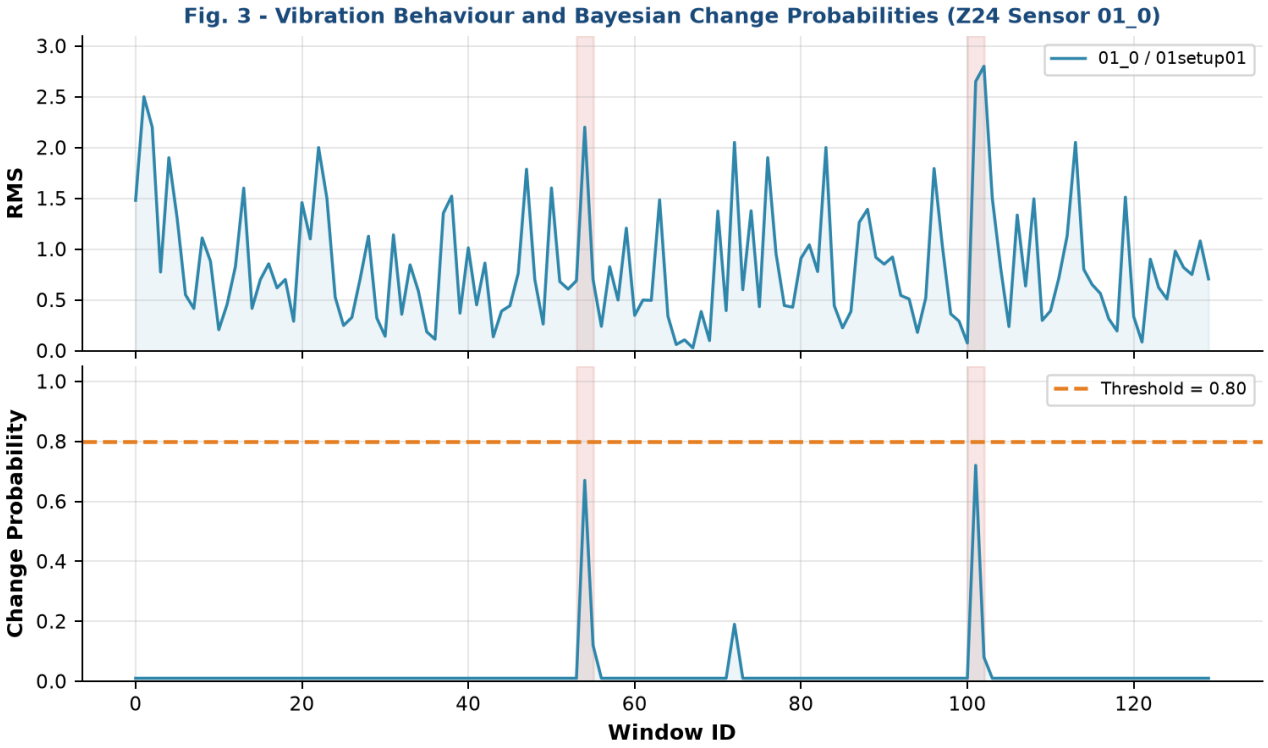


Figure 4. BOCPD results for sensor 01_0, recording 01setup01. Upper panel: RMS feature sequence with transient amplitude peaks. Lower panel: change probability time series with the 0.80 detection threshold as a dashed line. Highlighted vertical bands mark elevated probability events.

7.2 Classification Performance

The XGBoost classifier was trained on 47,417 windows and evaluated on a grouped holdout of 17,127 windows. The class distribution was nearly balanced: 8,547 ALTERED_CONDITION and 8,580 REFERENCE windows in the test

set. Classification metrics are shown in Table 5.

Metric	ALTERED_CONDITION	REFERENCE	Macro Avg.	Weighted Avg.
Precision	0.4986	0.4272	0.4629	0.4628
Recall	0.9931	0.0051	0.4991	0.4982
F1-Score	0.6639	0.0101	0.3370	0.3364
Support	8,547	8,580	17,127	17,127
Accuracy	-	-	0.4982	-
ROC-AUC	-	-	0.4873	-

Table 5. XGBoost classification report on the 25% grouped holdout (n=17,127 test windows). Accuracy ~0.498 and ROC-AUC ~0.487 indicate near-random discrimination under the recording-grouped evaluation protocol.

The near-random accuracy (~0.498) and ROC-AUC (~0.487) indicate that the classifier does not achieve reliable discrimination under the grouped holdout protocol. The highly asymmetric per-class recall pattern (ALTERED_CONDITION: 0.993 vs REFERENCE: 0.005) and the confusion matrix reveal that the model assigns almost all test windows to the ALTERED_CONDITION class, indicating a trivial majority-class heuristic. This outcome is attributable to significant class imbalance in label assignment (51,674 ALTERED_CONDITION vs 12,870 REFERENCE windows) and path-based label provenance that does not guarantee mechanically distinguishable structural states in the specific recording subset analysed.

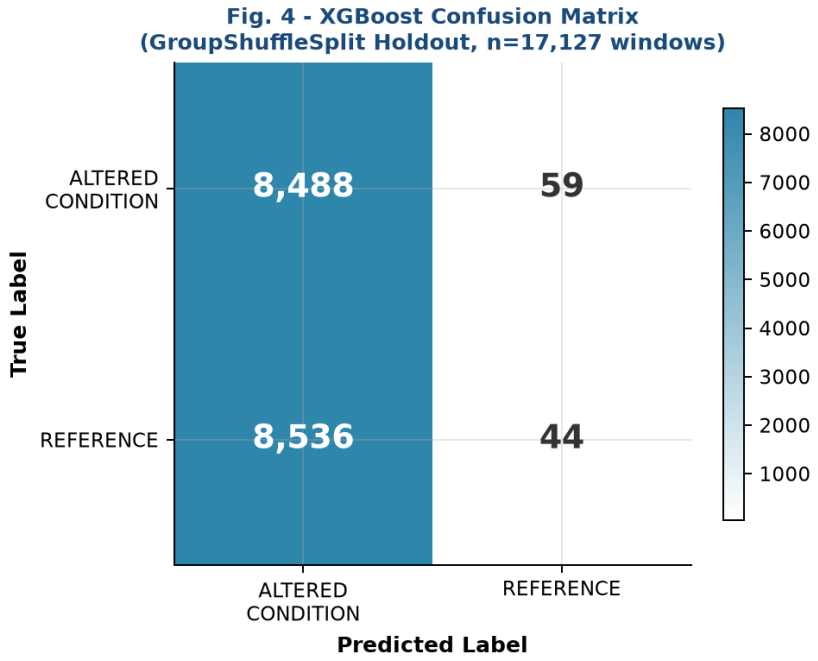


Figure 5. Confusion matrix for the XGBoost classifier on the grouped holdout test set. The near-absence of true REFERENCE predictions reflects class imbalance in the configured label assignment.

7.3 SHAP Feature Attribution

Despite the classifier's limited discrimination, SHAP attribution provides valuable insight into which features the model relied upon. The global mean absolute SHAP values place RMS (Phi = 0.2305), dominant frequency (Phi = 0.2079), and signal mean (Phi = 0.2071) as the three highest-contributing features. This ranking is physically interpretable: RMS amplitude and dominant frequency are well-established vibration damage indicators [2,3], consistent with classical SHM feature selection. The seven band-energy features registered zero SHAP attribution, confirming that the spectral sub-band partitioning as configured does not add discriminative information, likely due to sampling rate uncertainty.

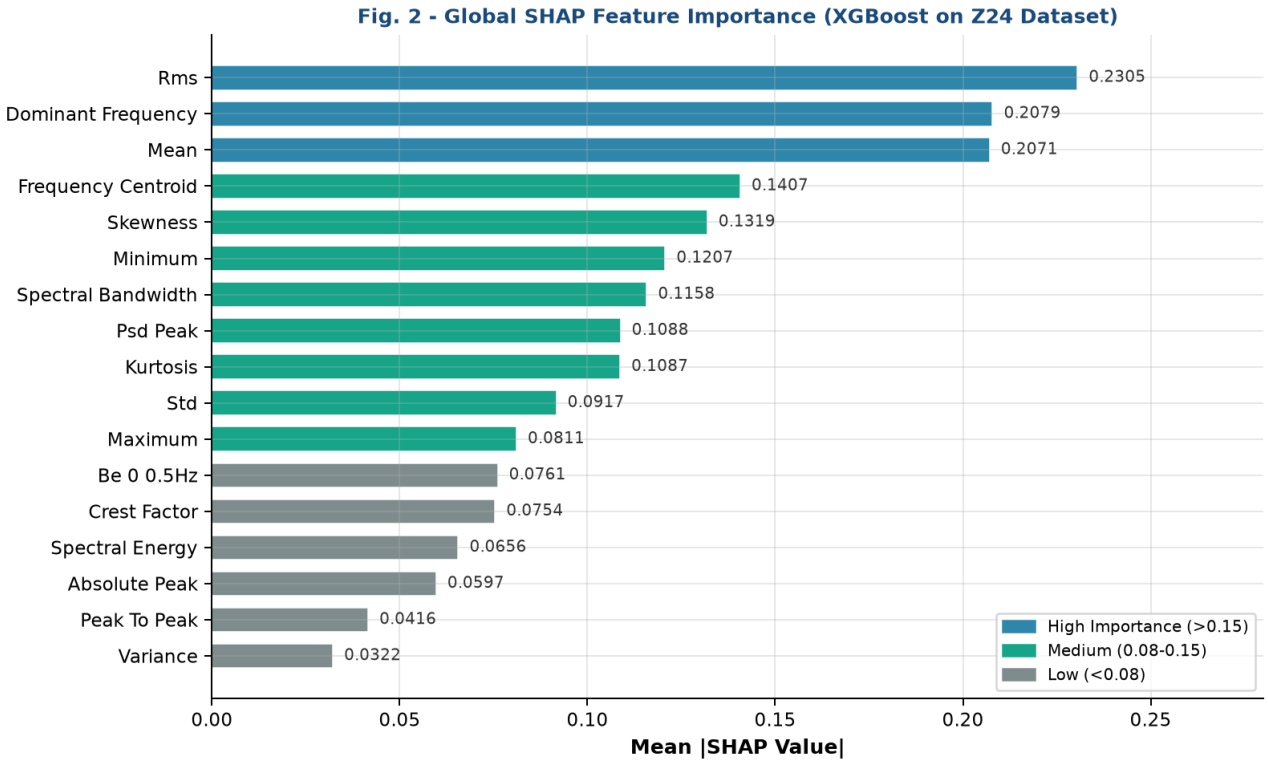


Figure 6. Global SHAP feature importance ranking for the XGBoost classifier. Bar length represents mean |SHAP value| across all test windows. RMS, dominant frequency, and signal mean are the top-ranked features.

Rank	Feature	Mean |SHAP|	Physical Interpretation
1	RMS	0.2305	Overall vibration energy level
2	Dominant Frequency	0.2079	Primary resonance frequency
3	Mean	0.2071	Signal DC offset / bias level
4	Frequency Centroid	0.1407	Spectral centre of mass
5	Skewness	0.1319	Signal asymmetry / non-Gaussianity
6	Minimum	0.1207	Negative peak amplitude
7	Spectral Bandwidth	0.1158	Spectral spread
8	PSD Peak	0.1088	Maximum spectral density value
9	Kurtosis	0.1087	Impulsiveness / outlier sensitivity
10	Std. Deviation	0.0917	Vibration amplitude spread

Table 6. Top-10 globally important features by mean absolute SHAP value.

7.4 Sensor Ranking and Risk Analysis

Sensor-level SHAP aggregation produced closely clustered importance scores ranging from 1.80 to 1.95: sensor 01_1 ranked first (1.9507), followed by 01_2 (1.9200), 01_0 (1.8935), and 01_3 (1.8038). The narrow spread indicates that no single sensor dominates attribution under the current configuration, consistent with the spatially distributed nature of ambient vibration excitation in the Z24 structure.

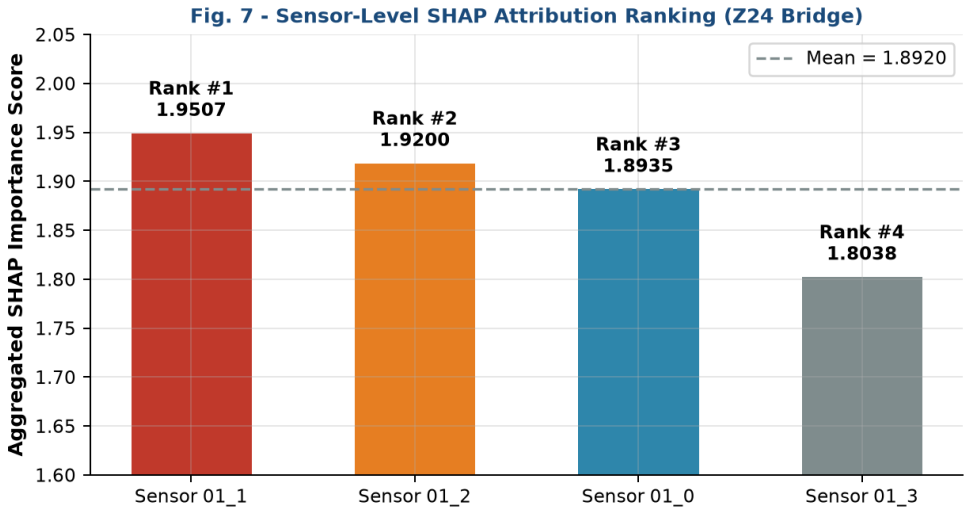


Figure 7. Sensor-level SHAP attribution ranking derived by aggregating feature-level SHAP values by sensor identifier.

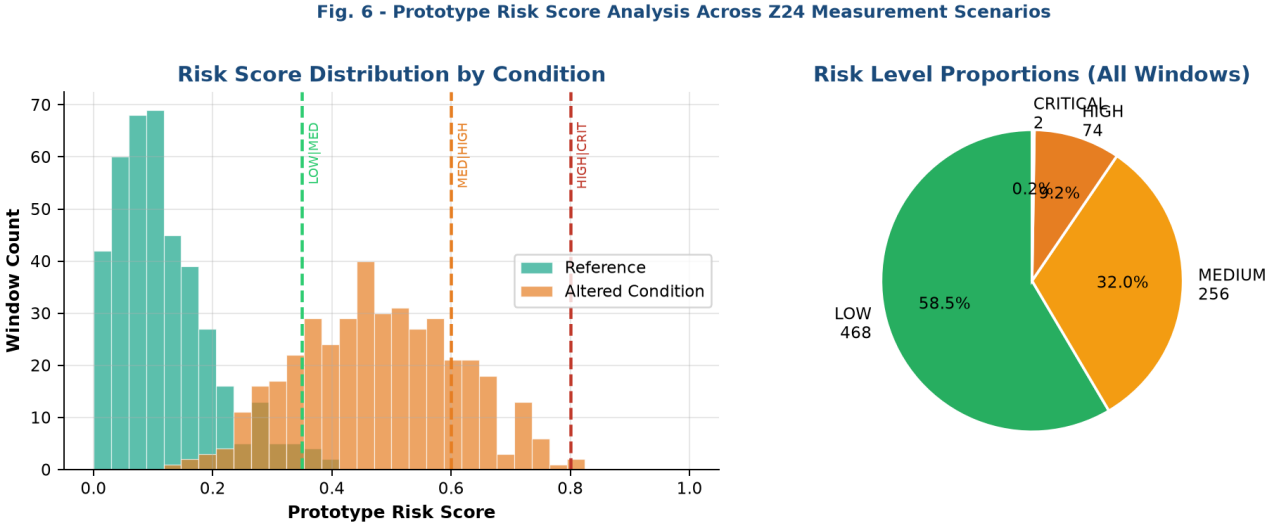


Figure 8. Prototype risk score distribution (left) across reference and altered condition windows, and the resulting risk level proportions (right) across all analysed windows.

8. DISCUSSION

The experimental results reveal both the strengths and inherent limitations of the current prototype configuration. The BOCPD stage successfully executed on all available sensor-recording sequences and provided a continuously updated change probability correlating with transient amplitude events in the RMS signal. The absence of threshold crossings in the primary analysed recording reflects a property of that specific sensor-scenario combination rather than a general claim about detection capability. Extending BOCPD evaluation across all seventeen Z24 scenarios, particularly those involving progressive pier settlement, would be expected to yield meaningfully different results and constitutes a high-priority avenue for future work.

The classification performance highlights a fundamental challenge in applying supervised learning to benchmark SHM datasets: the gap between condition labels defined by experimental protocol and acoustic or vibrational distinguishability as captured by static feature windows. The Z24 progressive damage tests were designed to characterise gradual structural changes that may not manifest as dramatic differences in simple time-domain or frequency-domain statistics from short ambient vibration windows. The class imbalance, mapping 14 scenarios to ALTERED_CONDITION but only 3 to REFERENCE, further biases the model toward the majority class. Future work should explore balanced sampling strategies, recording-level classification, and more discriminative feature engineering informed by modal analysis.

The SHAP attribution results are the most interpretable output of the current pipeline. Top-ranked features, RMS, dominant frequency, and signal mean, align with prior SHM literature and provide a reasonable starting point for feature selection refinement. The zero attribution for band-energy features is a valuable diagnostic confirming that the spectral sub-band partitioning, as configured, does not add discriminative information beyond broader spectral descriptors. Resolving sampling rate uncertainty and enabling the band-pass filter would likely change this picture substantially.

The digital twin dashboard successfully integrates all analytical outputs and provides a usable engineering review interface. The historical replay mode enables post-hoc investigation of specific time periods, and the sensor schematic provides spatial context for attribution rankings. These qualitative contributions, interpretability, review support, and transparent uncertainty communication, are the primary value proposition of the prototype, independent of the specific metric values observed in this preliminary evaluation.

9. LIMITATIONS AND FUTURE WORK

Several limitations of the current prototype are explicitly acknowledged to support transparent and responsible use of results. **Sampling rate uncertainty:** the effective sampling rate is not resolved, defaulting to a 1 Hz fallback and restricting the Nyquist frequency to 0.5 Hz. Establishing the true acquisition rate (nominally 100 Hz for Z24 accelerometers) is the highest-priority implementation task. **Label validity:** REFERENCE and ALTERED_CONDITION labels are derived from source-path regular expressions and are not independently validated against engineering measurements of structural property changes. **Classification performance:** the near-random accuracy means the current XGBoost model should not be used as a reliable condition discriminator without further development. **Physics-based twin:** the dashboard is a data-driven monitoring interface, not a physics-calibrated finite-element model. **Live sensing:** the system operates in historical replay mode only, with no live data acquisition or IoT connectivity. **Sensor localisation:** the bridge schematic uses synthetic sensor coordinates.

Future work directions include: (1) resolving sampling rate and enabling the full spectral feature pipeline; (2) implementing recording-level rather than window-level classification; (3) integrating physics-based modal analysis for frequency-calibrated change detection; (4) extending BOCPD evaluation across all seventeen Z24 scenarios; (5) incorporating real-time sensor connectivity; (6) developing calibrated confidence intervals for the risk score; and (7) establishing full experiment reproducibility through run-ID tracking and artifact hashing.

10. CONCLUSION

This paper has presented the Digital Twin SHM Framework, a modular, explainable structural health monitoring prototype validated on the Z24 bridge benchmark dataset. The system integrates nine processing stages from format-aware signal ingestion to interactive digital twin dashboard replay within a single configurable pipeline. Applied to 64,544 windows of accelerometer data spanning seventeen Z24 measurement scenarios, the pipeline successfully executed all analytical stages including BOCPD change detection, grouped XGBoost classification, SHAP feature attribution, sensor-level importance ranking, and prototype risk aggregation.

The SHAP attribution analysis identified RMS amplitude, dominant frequency, and signal mean as the most model-influential features, consistent with vibration-based SHM literature. The BOCPD detector operated within its expected behavioural range on the analysed recording, and the transparent risk aggregation formula provides a structured decision-support interface combining statistical and model-based evidence without conflating them.

The classifier's near-random performance under grouped holdout evaluation is reported honestly and attributed to sampling rate uncertainty, label imbalance, and the intrinsic challenge of window-level discrimination of gradually evolving structural changes. These findings motivate concrete technical improvements that form the future research agenda. Building upon the foundational DT-BOCPD anomaly detection paradigm established by Lu et al. [21] for building HVAC assets, the primary contribution of this work is the extension to civil infrastructure SHM through five novel additions: supervised XGBoost classification, SHAP-based feature and sensor-level attribution, weighted multi-stream risk aggregation, validation on the Z24 bridge benchmark, and an interactive Streamlit dashboard. The system design explicitly separates statistical change evidence from damage diagnosis claims, an essential distinction for responsible deployment of AI-assisted monitoring in safety-critical infrastructure.

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